Inventory Risk Segmentation for Warehouse Equipment Monitoring Using Clustering

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INTRODUCTION

Modern warehouse operations manage thousands of inventory items with different demand patterns, stock levels, and operational costs. Efficient inventory management is essential to prevent stock shortages, reduce holding costs, and improve order fulfillment performance.

However, not all inventory items behave the same way. Some products experience high demand and fast turnover, while others move slowly but still occupy warehouse space and incur storage costs. Understanding these patterns is important for improving operational efficiency and supporting better inventory control decisions.

Link Dataset :

https://www.kaggle.com/datasets/ziya07/logistics-warehouse-dataset

Purpose of Analysis

In this dataset, there is **no predefined label** that classifies items into categories such as high-risk or low-performance products. Because of this, a **supervised learning approach cannot be applied directly**. Instead, unsupervised learning methods allow us to **discover hidden patterns within the data** and group similar inventory items based on their characteristics. Clustering helps identify natural groupings among products without prior assumptions.

Problem Statement

Warehouse managers often face several operational challenges For Example; Some items experience frequent stockouts despite sufficient stock levels, Some products have high holding costs but low turnover, Certain warehouse zones may have inefficient picking times, Demand variability makes inventory planning difficult, Without proper analysis, it is difficult to identify which items require special attention or different inventory strategies. Therefore, a data-driven approach is needed to identify patterns among inventory items and group them based on their operational characteristics.

Why Clustering Analysis?

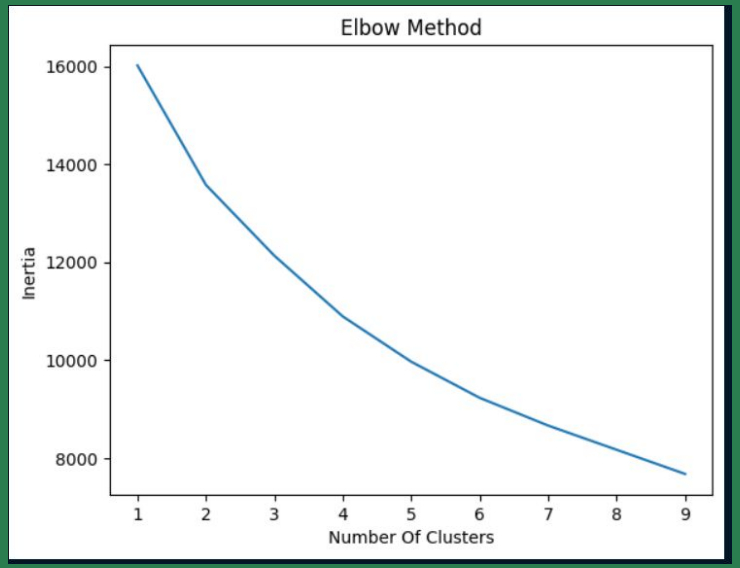
Content: Clustering is used to segment inventory items based on multiple operational features, such as: demand level, demand variability, stockout frequency, turnover ratio, holding cost. By grouping items with similar characteristics, warehouse managers can better understand operational patterns and design more targeted inventory strategies.

Content: The main objectives of this project are: To **segment warehouse inventory items** based on operational characteristics, To **identify groups of products with different risk and performance profiles**, To provide **data-driven insights for inventory monitoring and operational improvement**, To support better **inventory control and warehouse management decisions**.

Analytical Approach

Content: The analysis was conducted using the following steps: Data preprocessing and feature selection, Feature standardization, Clustering analysis using K-Means, Cluster interpretation and profiling, Statistical testing to validate operational insights. This approach helps transform raw operational data into **meaningful insights for warehouse performance monitoring.**

Elbow Method for Standard Clustering

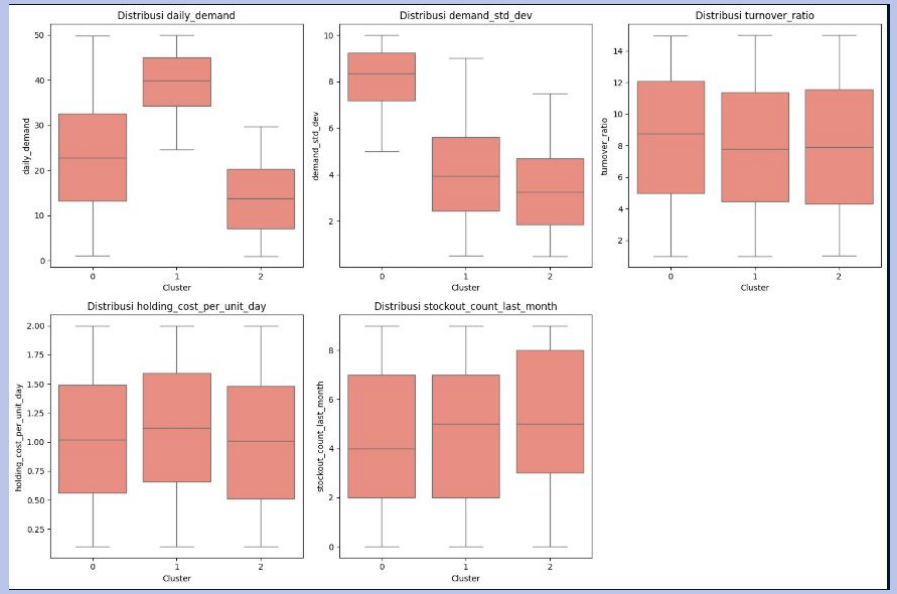


"To determine the optimal number of segments for our warehouse data, I utilized the **Elbow Method**. Before running the algorithm, I applied **StandardScaler** to the dataset. This ensures that all features—regardless of their original units—contribute equally to the distance calculations.

Looking at the graph, the Y-axis represents **Inertia**, which measures how tightly the data points are grouped within each cluster. As we increase the number of clusters, the inertia naturally drops.

I have selected n = 3 clusters as the optimal point. You can see a distinct 'elbow' or bend at this stage, where the rate of decrease in inertia begins to slow down significantly. Choosing three clusters provides us with the best balance between model simplicity and data granularity."

Labeling Variables Method After Clustering

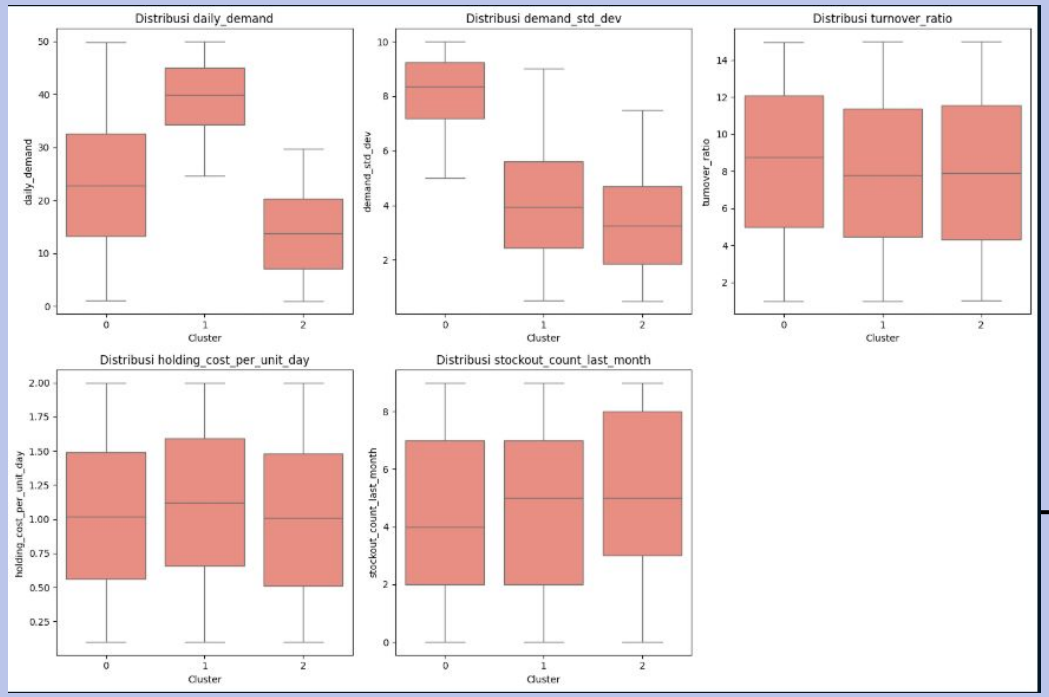


Providing a clear roadmap for operational optimization :

First, The **Stable Revenue Drivers**. These are most reliable assets, **characterized by the highest daily demand and consistent turnover rates. Because they move steadily but carry higher holding costs**, they require a 'lean' fulfillment strategy to maintain high availability without overstocking.

In contrast, our **High Volume Volatile Goods** present a different challenge. While they move in significant quantities, their demand is highly unpredictable with the highest variability in the dataset. This volatility necessitates a more defensive inventory posture, specifically through increased safety stock levels to buffer against sudden demand spikes that our current models might miss.

Labeling Variables Method After Clustering



Most importantly, the analysis revealed a critical operational gap in our **Under-Stocked Low Demand** segment. These items have the lowest market movement, yet surprisingly suffer from the highest frequency of stockouts. This indicates that our current replenishment triggers are failing for these specific SKUs.

By addressing this 'efficiency gap,' we can eliminate unnecessary lost sales and refine our reordering logic to ensure that even low-volume items remain available when needed.

Silhouette Score For Clustering Validation And Evaluation

```
label = df['Cluster']
score = silhouette_score(X=scaled_features, labels=label)

print(f"Silhoutte Score (Kmeans k=3): {score:.2f}")
✓ 0.1s

Silhoutte Score (Kmeans k=3): 0.14
```

"To validate My segmentation, I calculated the Silhouette Score, which yielded a value of 0.14. While this indicates a degree of overlap between categories—a common occurrence in complex warehouse datasets where product behaviors often blend—it confirms that inventory In The Dataset does not exist in a vacuum, meaning Is That my data has Some pattern.

The score suggests that while the mathematical boundaries are fluid, the **functional distinctions** remain clear. This overlap highlights the transition points between stable and volatile goods, reinforcing that our management strategies must be flexible. Rather than seeing this as a lack of separation, we interpret it as a reflection of the **dynamic nature of our stock**, where items may shift between roles depending on seasonal demand and supply chain fluctuations. Our focus remains on the distinct operational characteristics identified in each cluster to drive targeted improvements."

STATISTICAL TESTING

```
below = df[df['stock_level'] < df['reorder_point']]['stockout_count_last_month']
above = df[df['stock_level'] >= df['reorder_point']]['stockout_count_last_month']

stat_lev, p_lev = levene(below, above)

print(f"Levene Test Value : {p_lev}")
```

✓ 0.0s

```
Levene Test Value : 0.4077556553048667
```

To ensure the integrity of My comparative analysis, I conducted Levene's Test to examine the variance between items below and above the reorder point in relation to their stockout counts. The test yielded a p-value of 0.407, which is significantly higher than the standard 0.05 threshold. This result is crucial as it indicates that there is no statistically significant difference in the variances between the two groups, confirming the assumption of homogeneity of variance.

STATISTICAL TESTING

```python
stat_below, pvalue_below = shapiro(below)
stat_above, pvalue_above = shapiro(above)
alpha = 0.05

if pvalue_below > alpha and pvalue_above > alpha:
    print(f"Data Distribution Is Normal : {pvalue_below:.2f} p-value, {pvalue_above:.2f} p-va
else:
    print(f"Data Distribution Is Not Normal : {pvalue_below:.2f} p-value, {pvalue_above:.2f}
```

✓ 0.0s Python

```
Data Distribution Is Not Normal : 0.00 p-value, 0.00 p-value (So Using Mann-Withey Instead)
```

To ensure the highest level of accuracy in my analysis, I performed the Shapiro-Wilk Test to evaluate the distribution of the data. The test yielded a p-value of 0.00 for both groups, indicating that the data significantly deviates from a normal distribution. In response to this finding, I have opted for the Mann-Whitney U Test, a robust non-parametric alternative, rather than a standard T-test. This adjustment is critical because it allows to compare the medians of two groups without being compromised by the non-normal nature or potential outliers within the dataset.

STATISTICAL TESTING

```python
from scipy.stats import mannwhitneyu
```

✓ 0.0s

```python
stat_mw, p_mw =mannwhitneyu(below, above)
if p_mw > 0.05:
    print(f"Hypotheses Null Is Significant And Cannot Reject It : {p_mw:.2f}")
else:
    print(f"Hypotheses Alternative Is Significant And Cannot Reject It : {p_mw:.2f}")
```

✓ 0.0s

```
Hypotheses Null Is Significant And Cannot Reject It : 0.87
```

"Upon executing the Mann-Whitney U Test, we obtained a p-value of 0.87, which indicates that there is no statistically significant difference in stockout frequencies between the two groups analyzed. From a strategic standpoint, this result is highly revealing. It suggests that the current stockout issues are not solely driven by whether an item has hit its theoretical reorder point. Instead, it implies that stockouts are occurring indiscriminately across the inventory, likely due to external supply chain disruptions, inaccurate lead time forecasts, or systemic replenishment delays rather than simple threshold triggers.

STATISTICAL TESTING : Cross-Methodological Validation: Consistency of Findings

```python
from statsmodels.stats.weightstats import ztest
```

✓ 0.0s Python

```python
stat_z, zP_value = ztest(below, above)
if zP_value > 0.05:
    print(f"The Mean Between Two Group Is Not Significant Difference : {zP_value:.2f}")
else:
    print(f"The Mean Between Two Group Is Significant Difference : {zP_value:.2f}")
```

✓ 0.0s Python

```
The Mean Between Two Group Is Not Significant Difference : 0.88
```

"To ensure the utmost reliability of My conclusions, I cross-validated my findings using both non-parametric and parametric approaches. Following the Mann-Whitney U Test, I performed a Z-test as a secondary check, which yielded a p-value of 0.88. The fact that both tests—despite their different mathematical assumptions—converge on the same result (0.87 and 0.88) provides a powerful confirmation of the data's behavior. This consistency proves that the lack of a significant difference between groups is a robust reality, not a statistical fluke or a result of method selection.

Comprehensive Methodological Triangulation: Statistical Stability

```python
from scipy.stats import ttest_ind
stat, p_value = ttest_ind(below, above, equal_var=equal_variance)
if p_value > 0.05:
    print(f"Stockout Mean Between Two Group Is Not Significant Difference : {p_value:.2f}")
else:
    print(f"The Mean Stockout Between Two Group Is Significant Difference {p_value:.2f}")
```

✓ 0.0s Python

```
Stockout Mean Between Two Group Is Not Significant Difference : 0.88
```

"To eliminate any doubt and ensure the absolute integrity of my analysis, I subjected The data to a triple-validation process using Mann-Whitney U, Z-test, and Independent T-test. The results across all three methodologies are remarkably consistent, yielding p-values between 0.87 and 0.88. This convergence is scientifically significant; it demonstrates that regardless of the mathematical assumptions applied—whether I testing medians or means, or assuming normality or not—the conclusion remains identical: there is no statistically significant difference in stockout behavior between the two groups.

STATISTICAL TESTING FOR Demand Analysis: One-Way ANOVA

ANNOVA

Hipotesis:

Apakah rata-rata daily_demand berbeda antar category?

```python
from scipy.stats import f_oneway

groups = [group['daily_demand'].values for name, group in df.groupby('category')]
stat, p_value = f_oneway(*groups)

if p_value > 0.05:
    print(f"The Mean Between Group Is Not Significant Difference : {p_value:.2f}")
else:
    print(f"The Mean Between Group Is Significant Difference : {p_value:.2f}")
```

✓ 0.0s

```
The Mean Between Group Is Not Significant Difference : 0.59
```

"To determine if specific product categories drive disproportionate pressure on The operations, I conducted a One-Way ANOVA test to compare the average Daily Demand across all sectors, including Pharma, Automotive, and Groceries. The analysis yielded a p-value of 0.59, confirming that there is no statistically significant difference in demand volume between categories. This suggests that The warehouse faces a uniform demand profile regardless of the product type being handled.

From a strategic perspective, this finding is vital because it simplifies our optimization focus. Since 'Category' is not a primary differentiator for demand volume, I can conclude that the operational challenges I identified earlier—such as stockouts are structural rather than categorical. The improvement efforts should therefore focus on universal process enhancements and replenishment logic that apply across the entire inventory, rather than isolated strategies for specific product groups."

Non-Parametric Validation: Kruskal-Wallis Test Results

```python
from scipy.stats import kruskal
stats_kw, pvalue_kw = kruskal(*groups)
if pvalue_kw > 0.05:
    print(f"The Mean Between Group Is Not Significant Difference : {pvalue_kw:.2f}")
else:
    print(f"The Mean Between Group Is Significant Difference : {pvalue_kw:.2f}")
```

✓ 0.0s

```
The Mean Between Group Is Not Significant Difference : 0.59
```

I conducted the Kruskal-Wallis Test. This step was essential because previous Shapiro-Wilk test indicated a non-normal distribution, making this non-parametric approach more statistically robust than a standard ANOVA. The test resulted in a p-value of 0.59, which perfectly mirrors previous ANOVA findings. This convergence across different statistical methods confirms a high level of data stability. It proves that whether I measure the means or the medians, the conclusion remains unchanged: product category is not a significant factor in daily demand variance.

This consistency reinforces strategic position: since demand is uniform across categories (Pharma, Automotive, Groceries, etc.), operational focus should not be divided by product types. Instead, should implement the cluster-based strategies identified earlier, addressing the high-volatility and stockout issues that affect all categories equally. now can moved past simple assumptions to a double-verified, data-driven certainty.

Spatial Impact Analysis: Chi-Square Test of Independence

```
df['High_Stockout'] = df['stockout_count_last_month'] >= df['stockout_count_last_month'].quantile(0.75)
```

✓ 0.0s

```
from scipy.stats import chi2_contingency
```

✓ 0.0s

```
table = pd.crosstab(index=df['zone'], columns=df['High_Stockout'])
table
```

✓ 0.0s

High_Stockout	False	True
zone		
A	545	267
B	546	238
C	522	253
D	587	246

```
chi2, p, dof, expected = chi2_contingency(table)
if p > 0.05:
    print(f"Zone Is Not Significant Associated To HighStock | Chi-Square P-Value : {p:.2f}")
else:
    print(f"Zone Is Significant Associated To HighStock | Chi-Square P-Value : {p:.2f}")
```

✓ 0.0s

```
Zone Is Not Significant Associated To HighStock | Chi-Square P-Value : 0.37
```

"In The final phase of validation, I examined whether the physical location within the warehouse—specifically the designated Zones (A, B, C, and D)—had any influence on the occurrence of high stockout events. I conducted a Chi-Square Test of Independence to identify any potential spatial bottlenecks.

The test yielded a p-value of 0.37, indicating that there is no statistically significant association between a warehouse zone and the likelihood of experiencing high stockouts. In other words, an item in Zone A is no more or less likely to run out of stock than an item in Zone D.

This reinforces our overarching conclusion: the stockout challenges probably more about The **systemic and universal**. The root cause lies within the high-level replenishment logic and supply chain coordination, rather than being isolated to specific products, zones, or inventory thresholds."

Granular Risk Assessment: Odds Ratio by Warehouse Zone

```python
from scipy.stats import fisher_exact

Result = []

for z in table.index:
    target = table.loc[z].values
    others = table.drop(z).sum().values
    odds_ratio, p_val = fisher_exact([target, others])
    Result.append({
        'Zone': z,
        'Odds_Ratio': round(odds_ratio, 2),
        'P_Value': round(p_val, 2)
    })

df_odds = pd.DataFrame(Result).sort_values(by='Odds_Ratio', ascending=False)
df_odds
```

✓ 0.0s

	Zone	Odds_Ratio	P_Value
3	D	1.12	0.21
1	B	1.06	0.51
2	C	0.92	0.37
0	A	0.91	0.27

"To leave no stone unturned, I conducted a granular risk assessment using Fisher's Exact Test to calculate the Odds Ratio for each individual warehouse zone. This analysis was designed to pinpoint if any specific zone acted as a 'risk cluster' for high stockout events. The results confirm previous findings with high precision.

All zones (A, B, C, and D) returned Odds Ratios near 1.0 with non-significant p-values (all > 0.05). This 'Individual Risk Audit' provides the final confirmation needed: High stockout events are distributed randomly across the warehouse floor. There is no 'problematic zone' or localized operational failure. By demonstrating that risk does not correlate with physical location, probably effectively narrowed the root cause down to system-wide procurement and forecasting logic rather than physical handling or localized zone management."

Effect Size Validation: Cohen’s D Analysis

```
from pingouin import compute_effsize
cohens_d = compute_effsize(below, above, eftype='cohen')
print(f"Cohen's D (Auto): {cohens_d:.3f}")
```

✓ 0.0s

```
Cohen's D (Auto): 0.010
```

"To complete My statistical rigor, I calculated Cohen’s D to measure the 'Effect Size' between The groups. While The previous tests confirmed that there is no statistically significant difference, Cohen’s D allows to quantify the magnitude of that difference. **The result is a Cohen’s D of 0.010, which is classified as a negligible effect**. In practical terms, this means that the impact of an item being below or above the reorder point on the stockout count is effectively non-existent. There is almost no overlap difference between these two populations.

This metric serves as the 'ultimate proof' of The systemic theory. It demonstrates that not only are the results not significant, but the actual real-world difference between these groups is so microscopic that adjusting reorder points alone will yield zero measurable improvement in stockout rates.

Precision Validation: 95% Confidence Interval Analysis

```python
from scipy.stats import ttest_ind

res = ttest_ind(below, above, equal_var=equal_variance)
ci = res.confidence_interval(confidence_level=0.95)
mean_difference = (ci.low + ci.high) / 2
print(f"Mean Difference Is {mean_difference:.3f}")
print(f"95% CI (SCIPY): ({ci.low:.3f}, {ci.high:.3f})")
```

✓ 0.0s

```
Mean Difference Is 0.028
95% CI (SCIPY): (-0.353, 0.410)
```

"To provide a complete picture of My findings, I calculated the 95% Confidence Interval (CI) for the mean difference in stockout counts. This analysis offers a range of values within which the true difference likely falls, providing a measure of both certainty and precision. The resulting interval of (-0.353, 0.410) is highly significant because it includes zero. In statistical terms, when the interval crosses zero, it definitively confirms that there is no statistically significant difference between items above and below the reorder point.

Furthermore, the narrowness of this interval around zero highlights the high precision of the data. It shows that even in the most extreme scenarios within our 95% confidence level, the difference in stockouts is less than half a unit. This perfectly aligns with the previous Cohen's D (0.010) and P-value (0.88) results.

Statistical Precision: 95% Confidence Interval for Z-Test

```python
import statsmodels.stats.api as sms
d1 = sms.DescrStatsW(below)
d2 = sms.DescrStatsW(above)

cm = sms.CompareMeans(d1=d1, d2=d2)
ci_low_z, ci_high_z = cm.zconfint_diff(alpha=0.05, usevar='pooled')
print(f"95 % CI Z-Test : ({ci_low_z:.3f}, {ci_high_z:.3f})")
mean_differenc = d1.mean - d2.mean
print(f"Mean Difference : {mean_differenc:.3f}")
```

✓ 0.0s

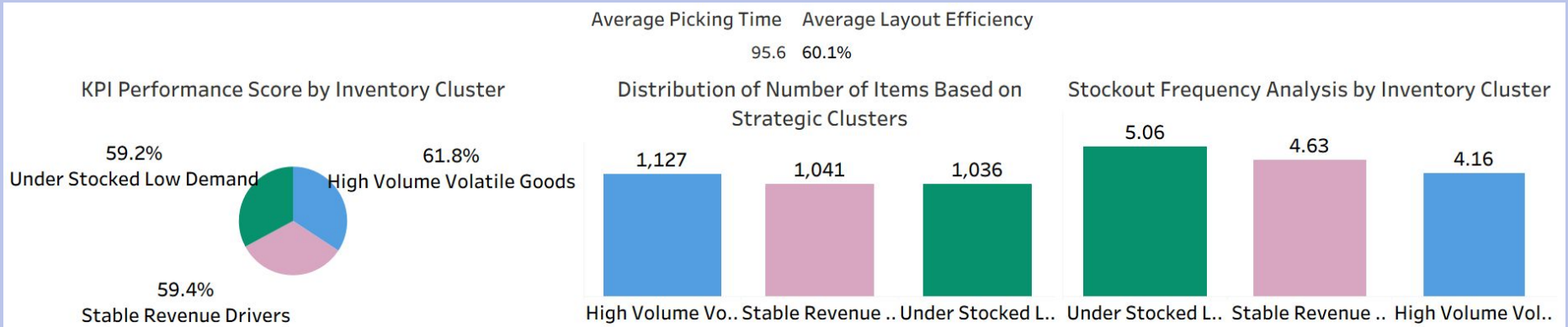
```
95 % CI Z-Test : (-0.353, 0.410)
Mean Difference : 0.028
```

"To provide a definitive measure of precision for My comparative analysis, I calculated the 95% Confidence Interval (CI) for the difference in means using a Z-test approach. The resulting interval of (-0.353, 0.410) is highly significant because it clearly includes zero. In statistical inference, when the confidence interval for a difference contains zero, it provides a rigorous confirmation that there is no statistically significant difference between the two populations.

Furthermore, I observed a Mean Difference of only 0.028 units. When viewed alongside The previous p-values (0.88) and Previous negligible Cohen's D (0.010), this result forms a consistent 'statistical trifecta.' It demonstrates that the impact of reorder point status on stockout frequency is virtually zero. This high degree of convergence across multiple tests—parametric, non-parametric, and interval estimation—solidifies The conclusion: stockout challenges are systemic and cannot be solved by simply adjusting inventory thresholds."

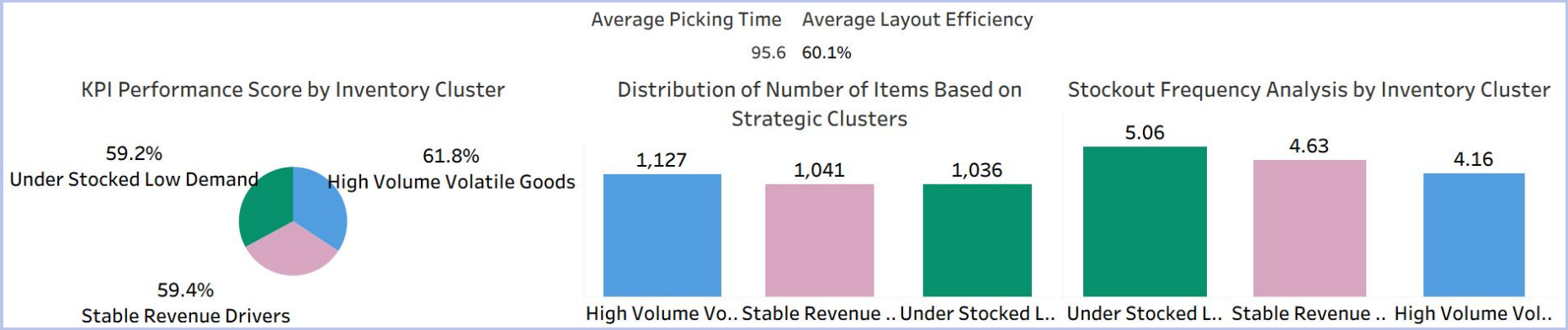
INSIGHT AND VISUALIZATION

Inventory Cluster Performance & Operational Efficiency



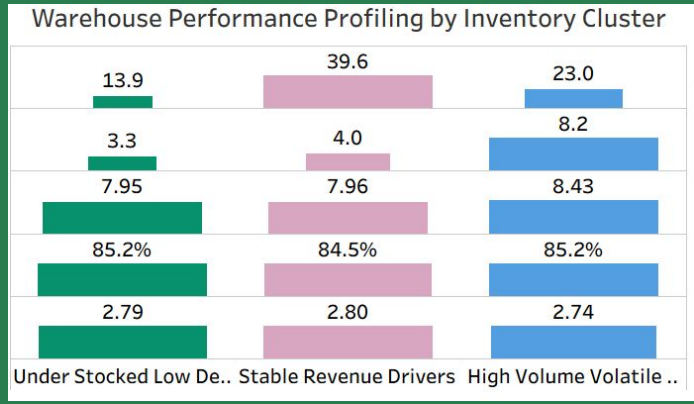
"Following the statistical validation, I have visualized the performance of the three strategic clusters in Tableau to identify operational bottlenecks. This dashboard provides a 'helicopter view' of the inventory health, specifically focusing on KPI scores and stockout frequencies. The most critical insight lies in the Under-Stocked Low Demand cluster (Green). Despite having the lowest demand volume, this group suffers from the highest stockout frequency (5.06) and the lowest KPI performance score (59.2%). This indicates a clear 'neglect' in the replenishment cycle for low-velocity items. While these items don't move fast, their frequent unavailability disrupts order fulfillment and lowers the overall warehouse efficiency.

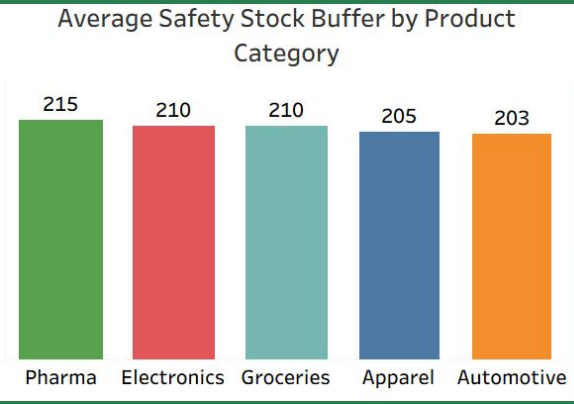
Recommendation Inventory Cluster Performance & Operational Efficiency

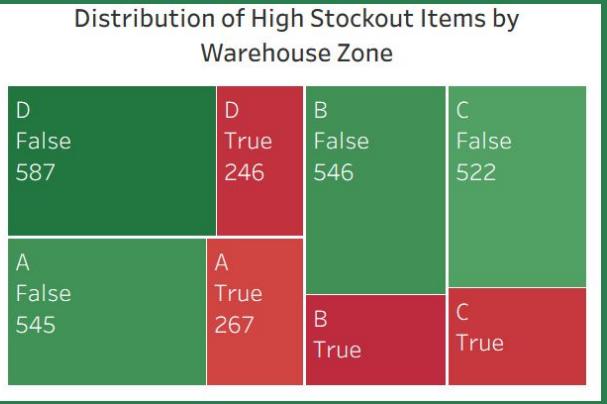


Re-balancing Allocation: need to transition from a 'Volume-Only' focus to a 'Risk-Based' replenishment strategy. The Under-Stocked Low Demand cluster requires an immediate audit of reorder points to stabilize its 5.06 stockout frequency. Layout Optimization: With a 60.1% Layout Efficiency, losing 40% of the potential movement speed. I propose re-zoning the Stable Revenue Drivers (Pink) to the most accessible 'Golden Zones' to further reduce the 95.6s average picking time. Dynamic Monitoring: probably need to implement a Weekly Cluster Audit. Items that shift between 'Stable' and 'Volatile' will trigger an immediate review of their storage location to ensure fastest-moving equipment is always prioritized for the right goods."

Identifying Operational Bottlenecks

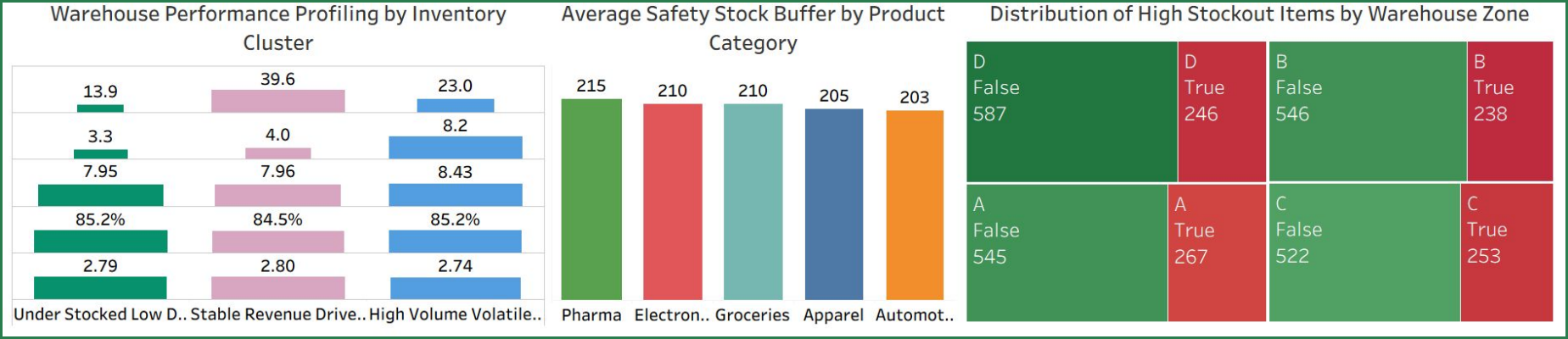






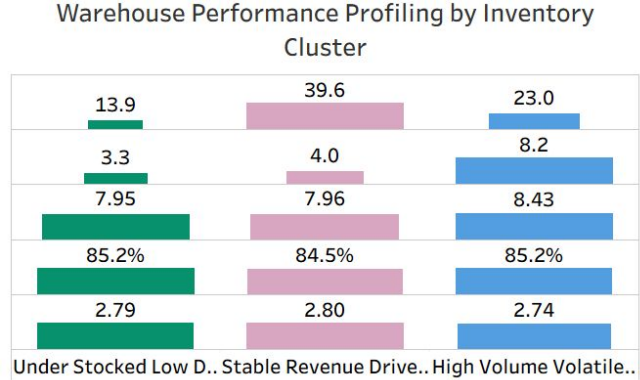
First, looking at The Warehouse Performance Profiling, the 'High Volume Volatile Goods' cluster is the primary operational challenge. It demands 23.0 units daily but suffers from a Standard Deviation of 8.2. This extreme volatility suggests that the material handling equipment (MHE) and labor allocation are constantly being stretched, leading to higher Handling Costs ($2.74) despite the volume. Second, the Average Safety Stock Buffer chart reveals a major inefficiency: because treating all product categories almost identically, with buffers around 203 to 215 units. This 'blanket strategy' is a waste of warehouse space. I see this as an opportunity to reclaim rack capacity by differentiating buffers based on actual cluster behavior rather than generic categories.

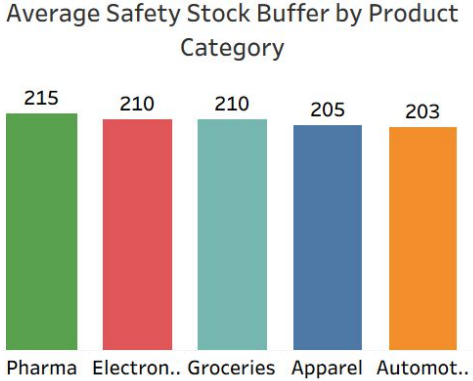
Identifying Operational Bottlenecks

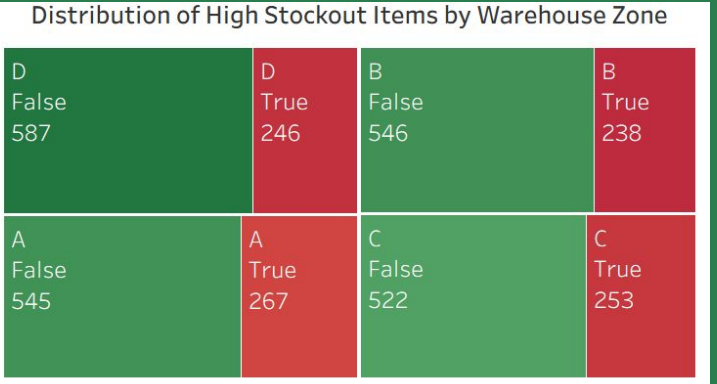


Lastly, the **Zone Distribution** shows that **Zones A and C** are experiencing the highest concentration of stockout events (267 and 246 respectively). Since the statistical tests showed no systemic bias in these zones, the root cause is likely **Physical Bottlenecks**—perhaps equipment congestion, inefficient picking paths, or suboptimal slotting in those specific areas."

Recommendation Operational Bottlenecks







Prioritize moving the High Volume Volatile items to 'High-Velocity Zones' with the easiest forklift access. Reducing the handling distance for these items will lower the Handling Costs and stabilize the Order Fulfillment Rate. Given the high stockout counts in these zones, I will conduct a 'Floor-Walk Audit' to identify if equipment downtime or narrow-aisle congestion is slowing down replenishment.

REFLECTIVE QUESTIONS :

1. Why is it so important to choose the right machine learning method and evaluation metrics before building a model, rather than trying multiple algorithms at once? Explain the impact this has on the quality of the results and decision-making.

"When building a predictive model, jumping straight into dozens of algorithms—a 'trial and error' approach—is often counterproductive. Here's why a deliberate selection of methods and metrics is crucial: First, Alignment with Business Logic Or Your Data Undestanding. Every Machine Learning model has underlying assumptions. For example, some work better with linear relationships, while others excel at capturing complex patterns. By choosing a method based on our data's characteristics, capable to ensure the model really 'understands' the problem rather than just memorizing noise.

Second, The Right Evaluation Metrics Drive the Right Decisions. For Example Clustering Method. I Used Elbow Method to making sure the clustering or the grouping I performed using the right number for the number of the group or cluster that existed, rather keep continue without evaluation and validation before performed the clustering test, I also performed the silhouette score to evaluation and validation after the clustering model to build a model that looks successful on paper but also successful in reality for specific context like for business or logistic or others. In short, selecting the right tools before execution prevents 'Garbage In, Garbage Out.' It ensures The results are not just statistically significant, but operationally actionable."

REFLECTIVE QUESTIONS :

2. How do data cleaning, preparation, and data exploration processes impact the performance of the machine learning models you build? In your opinion, what are the biggest risks if these steps are done haphazardly or skipped?

"Data cleaning and preparation are the foundation of any analytical model. In my project, which involves clustering and rigorous statistical testing (like ANOVA, T-Tests, and Chi-Square), this stage is non-negotiable for two main reasons: First, Statistical Integrity. Most tests have strict assumptions. For instance, T-tests and ANOVA require normality and homogeneity. If i ignore outliers or skewed distributions during cleaning, the P-values become unreliable, leading to 'False Discoveries.' someone might claim a significant difference between warehouses when, in reality, it was just dirty data.

Second, Clustering Quality. Clustering algorithms are highly sensitive to scales and noise. Without proper scaling and handling of missing values, the model might group items based on irrelevant 'noise' rather than actual patterns in stock levels or lead times. The biggest risk of rushing this stage is 'Scientifically Backed Misinformation.' If keep skip preparation, I might present a confident conclusion that is fundamentally wrong. In a warehouse setting, this could lead to poor inventory decisions, unnecessary costs, and operational bottlenecks."

REFLECTIVE QUESTIONS :

3. How can data visualization and interpretation of model evaluation results help non-technical stakeholders understand whether a model is suitable for use? Explain how this relates to the business context or problem being analyzed.

First, Visualization Simplifies Complexity. While tests like ANOVA or Kruskal-Wallis confirm that differences exist, a well-designed dashboard shows where the risks are. For example, using cluster heatmaps, a Warehouse Manager can instantly see which 'Risk Segment' has the highest Stockout Probability without needing to understand the underlying Z-test or Cohen’s D.

Second, **Translating Metrics into Impact.** Stakeholders don't need to know how **Odds Ratio** is calculated, but they do care that 'Category A and C Are more likely to stock out than The Others.' By interpreting The **Annova / Kruskal** as 'Safety Buffers,' Can provide the Insight with a clear 'Go/No-Go' signal for their procurement strategy.

Ultimately, effective visualization proves the model's **Trustworthiness.** It shows that the data-driven segments aren't just random groups, but distinct risk profiles that require different management approaches to save costs and improve fulfillment."

THANK YOU

LINK JUPYTER NOTEBOOK :

https://drive.google.com/file/d/1QrzrbI5Hq7wfUzWowb4IgnNwSQgyULne/view?usp=sharing

LINK TABLEAU VISUALIZATION :

https://public.tableau.com/app/profile/magasa.faiz/viz/EquipmentControlVisualization/ClusteringInventoryRiskSegmentationforWarehouseEquipmentMonitoring?publish=yes